

## Education

University of Waterloo - BAsC Electrical Engineering (Honours, Coop), 4A, Expected Graduation: April 2027

## Skills

Languages: C++, C, Python, VHDL

Embedded & Systems: FreeRTOS, POSIX, MQTT, CAN, STM32, BSP, VFS

Tools & Development: VS Code, Makefiles, OpenOCD, GDB, STM32CubeIDE, Docker, KiCad, Git, MATLAB

Certifications: Six Sigma Green Belt (40 hrs), Internal Combustion Engine training (40 hrs)

## Work Experience

AlumaPower Corporation - Firmware Engineer Jan 2026 - April 2026

- Implemented SNTP based time synchronization using a Chrony server, modifying FreeRTOS POSIX layers and W6100 network drivers to provide system wide POSIX timekeeping.
- Authored POSIX style ioctl drivers for MCF8316C, INA745, TCA9535PWR, TCA9548 and MCP9843 devices with VFS integration.
- Contributed OS level enhancements within the BSP layer, implementing VFS path construction, service threads, PHY link handling and resolving system stability bugs.
- Developed and debugged application layer firmware across multiple distributed subsystems.
- Authored Doxygen documentation for firmware drivers and OS interfaces to improve maintainability.
- Developed Node-RED dashboards and flows for MQTT telemetry visualization and subsystem control.

AlumaPower Corporation - Firmware Engineer Jan 2025 - April 2025

- Developed and debugged embedded C firmware on STM32G474RE using Makefiles and OpenOCD, implementing FreeRTOS state machines for motor controller subsystems with MCF8316C controllers.
- Designed POSIX style I2C drivers for the MCF8316C motor controller, including EEPROM encode/decode layers and high-level control wrappers.
- Contributed to board support package (BSP) abstractions for flexible peripheral configuration (I2C, CAN, EEPROM) across subsystem boards.
- Built a Docker based windows development environment with standardized compilers, libraries and OpenOCD integration for consistent cross laptop firmware debugging.
- Developed Python test tools for CAN and MQTT based subsystem communication, enabling telemetry monitoring, CSV logging and manual/automated testing.
- Designed and assembled a remote debugging PCB with Raspberry Pi CM4 and STLINK-V3MODS for GDB/OpenOCD access without direct hardware connection.

AlumaPower Corporation - Systems Technician May 2024 - August 2024

- Developed FreeRTOS firmware in C for STM32G474RE controlled fluid management system.
- Implemented CAN TX/RX, interrupt driven timers, a multi state control state machine which included 1-Wire temperature sensors, level sensors and flow sensors for event triggering.
- Designed and built a hardware safety shutdown circuit using DPDT relays, integrating readings from hydrogen, airflow and fluid level sensors.
- Created python tools for CAN message filtering and decoding over MQTT, enabling real time debugging and telemetry visualization.

FORD Motor Company - R&D Test Engineer September 2023 - December 2023

- Conducted engine failure analysis and vibration testing using accelerometers and MATLAB.
- Managed engine dyno cells and developed VBA/SharePoint tools to automate calibration workflows.
- Designed and built an engine test bench using wiring harnesses to safely test individual engine components.

Clark Construction Management - Field Engineer January 2023 - April 2023

- Interpreted structural drawings to quantify material requirements and coordinate large scale concrete pours (~6,400 m<sup>3</sup>), managing field sequencing and supplier logistics.

JC & J Byrne Farms 2017 - Present

- Maintain and troubleshoot mechanical and electrical equipment in agricultural operations.